The University of Nottingham Ningbo China

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, Autumn SEMESTER 2023-2024

Database and Interfaces

Time allowed: **TWO Hours**

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer ALL Questions.
There are 50 marks in total.

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn your examination paper over until instructed to do so

ADDITIONAL MATERIAL: None.

INFORMATION FOR INVIGILATORS: Exam papers must be collected at the end of the exam.

Question ONE – Web Programming [15 marks]

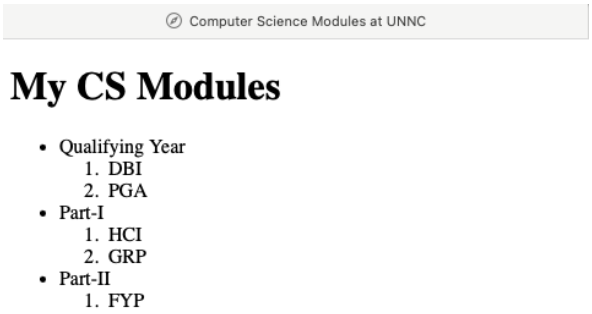
a) What is the application layer protocol that powers the world wide web and allows for retrieving web pages and other files from web servers?

[1 mark]

b) What are HTML attributes and how are they used within HTML element tags? Provide an example of an HTML attribute properly used within a specific tag to support your explanation.

[2 marks]

c) Write the full HTML document structure necessary to represent the webpage shown in the following figure. The text "My CS Modules" is a level-1 heading. There is no need to include any CSS rules in your solution. Note that you **should include** the page title, as shown in the figure.



[5 marks]

d) The following HTML code specifies two inputs for name (text) and age (number), along with a submit button:

```
<form action="process_form" method="post">
  <label for="name">What is your name?</label>
  <input type="text" id="name" name="user_name">
  <label for="age">What is your age?</label>
  <input type="number" id="age" name="user_age">
  <input type="submit" value="Submit">
</form>
```

Using Flask, write a function that:

- Retrieves the user's name and age input from the form.
- Validates that the age input, converted to an integer, is 18 or over.
- Returns a simple text response based on the following input conditions:
  - If age is under 18, return "You must be over 18!"
  - If name is not empty and age is >=18, return "Hello X", where X is the name
  - If name is empty and age is >= 18, return "Hello there!"
  - If both fields are empty, return "How old are you?"
- Only handle the specified input combinations, no need to consider other cases.
- The return value should be a plain text string, no HTML required.

[7 marks]

Question TWO – Relational DB Design Theory and DBMS [15 marks]

Consider the following relations:

Customer

cID	cName
1	Tom
2	John
3	Mike

Purchase

cID	bCode	pItem	pCost
1	ROG	Graphic Card	18000
1	PGA	Memory	800
1	ROG	Monitor	5500
2	MSI	Graphic Card	14500
2	COR	Memory	350
3	GALAX	Graphic Card	15000
3	KT	Memory	600
3	HP	Monitor	3000

Brand

bCode	bName
MSI	Micro-Star International
ROG	Republic of Gamers
GALAX	Galaxy Microsystems
GS	G.Skill
COR	Corsair Gaming
KT	Kingston
HP	Hewlett-Packard

a) Write down the results of the following relational algebra expressions. If an expression is not valid, please explain the reason why.

i. $\sigma_{\text{cID = 1 and price > 5000}}(\text{Purchase})$

[1 mark]

ii. $\pi_{\text{bName}}(\text{Brand} \bowtie \sigma_{\text{price>2000}}(\text{Purchase}))$

[2 marks]

iii. $\pi_{\text{cName}}(\text{Customer} \bowtie \sigma_{\text{pItem = "Memory" and pCost<500}}(\text{Purchase})) \cup \pi_{\text{cName}}(\text{Customer} \bowtie \sigma_{\text{pItem = "Graphic Card" and pCost>15000}}(\text{Purchase}))$

[3 marks]

b) List the output of the following SQL query:

```
SELECT DISTINCT cName FROM Customer NATURAL JOIN Purchase
WHERE cID NOT IN (SELECT cID FROM Purchase WHERE pItem = 'Monitor');
```

[2 marks]

c) Write an SQL query that returns the customer's name, brand code, and total spend for the brand they spent the most money on for each customer that has made at least one purchase. Make sure to exclude any customers who do not have any purchases and return only one row per customer showing the details for their highest spend brand.

[4 marks]

d) What is a database transaction? What do transactions guarantee? When should they be used?

[3 marks]

Question THREE – SQL and ER Design [20 marks]

You have been asked to design a database for the Software Engineering Group Project (GRP) module. In this module, students are divided into teams, with each team working to develop and document a large software engineering project. Each team will be supervised by one teacher.

- A teacher has a unique staff ID (an integer), a name and gender.
- A student has a unique student ID (an integer), a name, and gender.
- A team is formed of two or more students. A student can only belong to one team.
- Each team has its own unique ID (an integer) and a team name.
- Each teacher will provide exactly one project topic that can be allocated to a selected team. A topic will consist of a title (text only) and project description (text only).
- Each teacher selects which team will be allocated his/her topic. A team may only be allocated one topic. A given topic may only be allocated to one team.
- Each team's performance is assessed by exactly three teachers. Each teacher will assess several team projects.

Note, it is expected that you make reasonable assumptions, based on your real-life experiences, in the design and implementation of this database. Assumptions should be written down.

a) Draw the Entity Relationship Diagram (ERD) for the GRP module. Please draw all entities, attributes in entities, and the cardinalities between entities.

[8 marks]

b) Write the necessary SQL code to represent your database design in a modern DBMS system. Be certain to indicate primary keys and referential integrity constraints. Sensible referential integrity constraints should be assumed if they are not clearly described in the problem description.

[8 marks]

c) Please write the necessary SQL to list all IDs of students who had an average team score greater or equals to 70.

[4 marks]